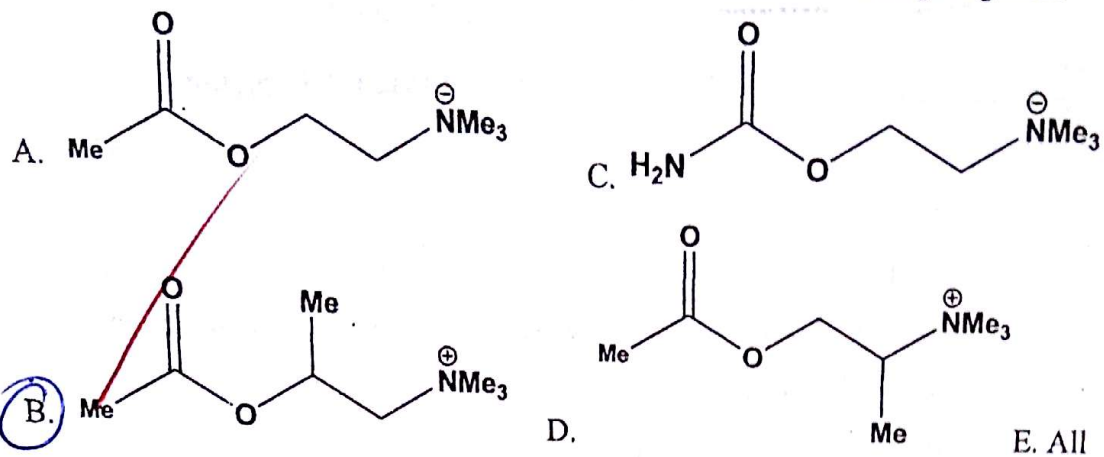


I. Choose the best answer and Encircle your Letter of choiceName Chachu Genale ID No HSE/0818/08

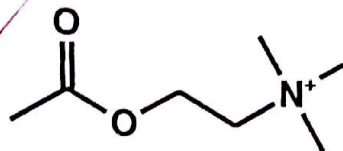
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1. One of the following adrenergic agonist is norepinephrine reuptake inhibitor and facilitate the release of norepinephrine
☒ A. Ephedrine ☒ B. Cocaine ☐ C. Desimpramine ☐ D. Amphetamine ☐ E. Tyramine
2. Accidental discovery of drugs in the laboratories or clinics by pharmacists, physicians or other investigators is known as
A. Rational approach ☒ C. Serendipity
B. Random screening D. Rational directed screening E. None
3. All of the following are true except one
A. α adrenergic agonists used to treat arrhythmia
☒ B. β adrenergic agonists used to treat glaucoma
C. α adrenergic agonists used as antihypertensive
D. β adrenergic agonists used as cardiac stimulant after surgery E. None
4. One of the following nomenclature of drugs is chosen and given by official agencies
A. Chemical name ☒ C. Brand name
B. Code name D. Proprietary name ☒ E. Nonproprietary name
5. All of the following drugs prevent release of acetylcholine from synapse except one
A. Vesamicol ☒ C. Botox
☒ B. Physostigmine D. Hemicholinium chloride E. None
6. In molecular modification of drug designing the association of different moieties through weak forces such as electrostatic attraction and hydrogen bonding is
A. Molecular Hybridization ☒ C. Molecular Addition
B. Molecular Replication D. Molecular Simplification E. All
7. One of the following drug is antidote for Acetylcholine Esterase Inhibitors (pesticide or nerve gas) poisoning
☒ A. Pralidoxime chloride B. Hydroxylamine C. Atropine ☒ D. Parathion E. None
8. One of the following is not reversible inhibitor of Acetylcholinesterase
A. Neostigmine ☒ B. Donepezil C. Miotine ☒ D. Isoflurophate E. None
9. One of the following parameter does not influence drugs activity
A. Lipophilicity of the drug C. Electronic factors of the drug
B. Size of the drug D. Shape of the drug ☒ E. None

10. Which one of the following is muscarinic receptor selective cholinergic agonist?

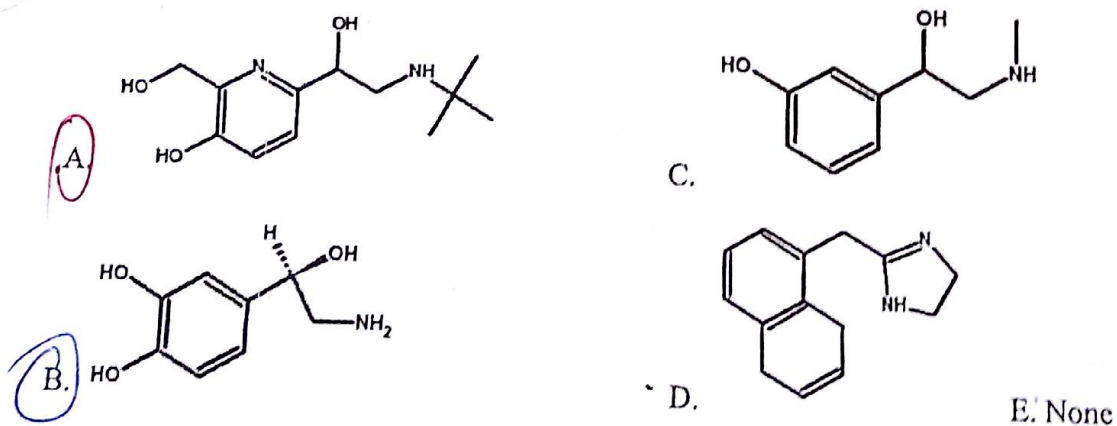


11. All of the following are true about the SAR of acetylcholine, except



- A. Quaternary nitrogen is essential
 B. Methyl group of acetoxy group can be extended
 C. Substitutions at α and β with larger groups than methyl decrease activity
 D. Ester is important E. None

12. Which one of the following adrenergic agonist is β receptor selective?



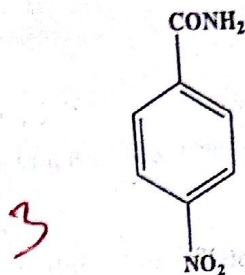
13. All of the following are β adrenergic antagonists except one, which one is the exception?

- A. Pronethalol C. Esmolol
 B. Propranolol D. Prazocine E. None

14. Which one of the following is the source for drug discovery?

- A. Plants B. Animals C. Minerals D. Synthetic sources E. All

4. Calculate the lipophilic substituent constant of benzamide in para nitro benzamide if
logP value of benzene = 2.13, logP value of p-Nitrobenzamide = 1.65 and $\pi_{\text{NO}_2} = 0.86$
(3pts)



~~$\pi_{\text{NO}_2} = \log P$~~

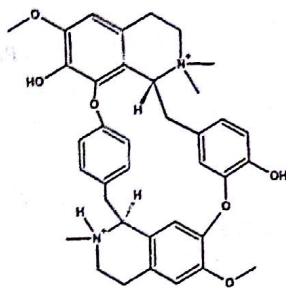
$$\pi_{\text{CONH}_2} + \pi_{\text{NO}_2} = \log P_{\text{Nitrobenzamide}} - \log P_{\text{benzene}}$$

$$\pi_{\text{CONH}_2} + 0.86 = 1.65 - 2.13$$

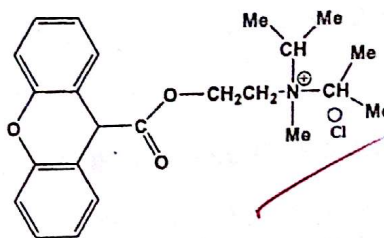
$$\pi_{\text{CONH}_2} = 1.65 - 2.13 - 0.86$$

$$= -1.34$$

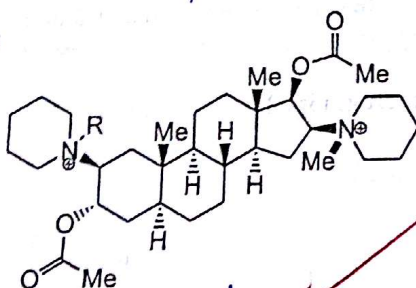
5. Classify the following acetylcholine antagonists as Muscarinic and Nicotinic antagonist (4pts)



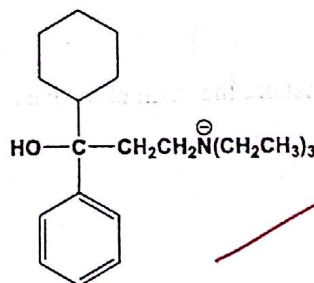
a Nicotinic



b Muscarinic

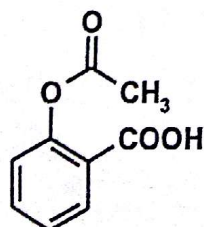


c Nicotinic



d Muscarinic

6. Does Lipinski's "rule of five" predict good oral bioavailability for Aspirin? Evaluate it by using Lipinski's "rule of five" (3 Pts)

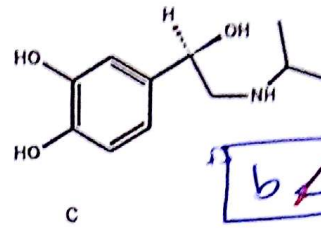
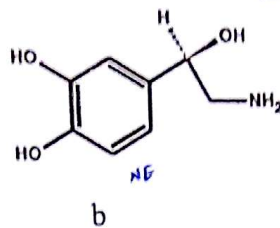
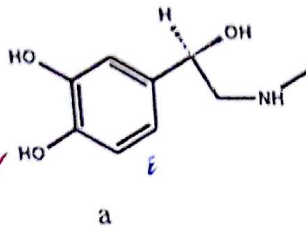


1. MW < 500
2. H-bond donating group ~~2~~ = 1 (< 5)
3. H-bond accepting group = 3 (< 10)
4. clogP b/w -1 and +5

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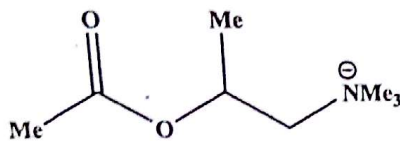
=> Aspirin - good oral BA

7. Arrange the following adrenergic agonists in increasing order of β -selectivity (2pts)

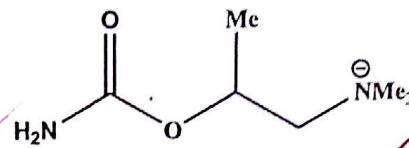


b < a < c

8. Compare briefly with reasons which one of the following cholinergic agonists is stable to acid hydrolysis and esterase enzymes. Give the generic name when requested (3 pts).



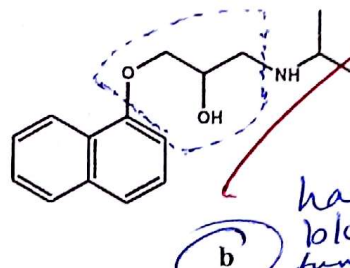
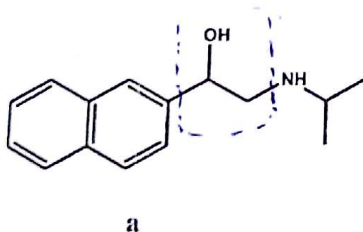
Generic Name... Methacholine



Generic Name... Bethanecol

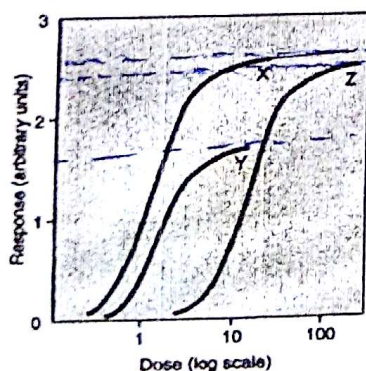
✓ is more stable because it has both electronic (H₂N-) and steric (Me) hindrances

9. Explain briefly with reason which of the following β -adrenergic antagonist has greater activity than the other (2pts)



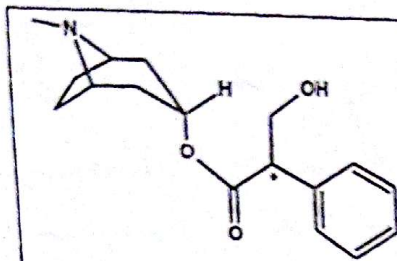
has greater activity, b/c the enclosed functional group.

10. Compare the potency and efficacy of x, y and z in the following hypothetical dose-response curve (4pts)



potency: $x > y > z$
Efficacy: $x \geq z > y$

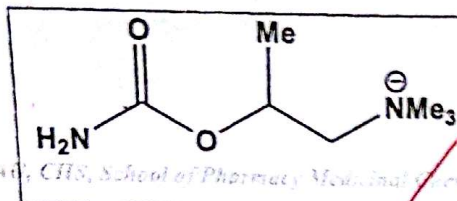
15. True about the drug illustrated below:



- i. Decreases GIT motility
- ii. Antidote for anticholinesterase poisoning
- iii. Treatment of Parkinson's Disease
- iv. Cholinergic agonist

A. i and iii only B. i, ii and iv only C. i and ii only D. ii and iii only **E. i, ii and iii only**

16. True about the drug illustrated below



- i. Muscarinic receptor selective agonist
- ii. Is more stable than Carbachol and Methacholine
- iii. Nicotinic receptor selective agonist
- iv. To treat urinary retention and abdominal distension

15. True about the drug illustrated below:

A. i, ii and iii only **B. i, ii and iv only** C. i and ii only D. i and iii only E. All

II. Answer the following questions briefly in the space provided

1. Why do we need new drugs (3 pts)?

- ✓ TO combat drug resistance
- ✓ TO come up with more effective drugs
- ✓ TO treat newly existing diseases. (outbreaks), etc.

A. i and iii only B. i, ii and iv only C. i and ii only D. i and iii only E. All

2. Write at least 4 routes of new drug or lead structure discovery (3pts)

- ✓ Serendipity (accidental discovery)
- ✓ Clinical trials
- ✓ Experimental trials
- ✓ Random screening
- ✓ High throughput screening
- ✓ Rational approach
- ✓ Rational directed screening

3. What are the main objectives of designing Prodrugs (3pts)?

- ✓ To improve pharmacokinetic and P'dynamic properties
- ✓ In order to make drugs acceptable by patients. (For instance by adding flavorants)
- ✓ To make the drug to be released in the target site (= site specificity) etc.